

Marriage and life satisfaction: T-Test

Notes

Output Created		06-NOV-2024 15:57:53
Comments		
Input	Data	D:\survey.sav
	Active Dataset	DataSet2
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	N of Rows in Working Data File	439
Missing Value Handling	Definition of Missing	User defined missing values are treated as missing.
	Cases Used	Statistics for each analysis are based on the cases with no missing or out-of-range data for any variable in the analysis.
Syntax		T-TEST GROUPS=Martial_grouped (1 2) /MISSING=ANALYSIS /VARIABLES=tlifesat /ES DISPLAY(TRUE) /CRITERIA=CI(.95).
Resources	Processor Time	00:00:00.02
	Elapsed Time	00:00:00.01

Group Statistics

	married & remarried	N	Mean	Std. Deviation	Std. Error Mean
Total life satisfaction	1.00	218	23.57	6.676	.452
	2.00	218	21.19	6.667	.452

Independent Samples Test

		Levene's Test for Equality of Variances		t-test for Equality of Means	
		F	Sig.	t	df
Total life satisfaction	Equal variances assumed	.039	.844	3.726	434
	Equal variances not assumed			3.726	433.999

Independent Samples Test

		t-test for Equality of Means		
		Significance		Mean Difference
		One-Sided p	Two-Sided p	
Total life satisfaction	Equal variances assumed	<.001	<.001	2.381
	Equal variances not assumed	<.001	<.001	2.381

Independent Samples Test

		t-test for Equality of Means		
		Std. Error Difference	95% Confidence Interval of the Difference	
			Lower	Upper
Total life satisfaction	Equal variances assumed	.639	1.125	3.637
	Equal variances not assumed	.639	1.125	3.637

Independent Samples Effect Sizes

				95% Confidence Interval	
Standardizer ^a			Point Estimate	Lower	Upper
Total life satisfaction	Cohen's d	6.671	.357	.167	.546
	Hedges' correction	6.683	.356	.167	.545
	Glass's delta	6.667	.357	.166	.547

a. The denominator used in estimating the effect sizes.

Cohen's d uses the pooled standard deviation.

Hedges' correction uses the pooled standard deviation, plus a correction factor.

Glass's delta uses the sample standard deviation of the control (i.e., the second) group.

Question2: Marriage and optimism levels: One-way Anova

Notes

Output Created		06-NOV-2024 16:48:31
Comments		
Input	Data	D:\survey.sav
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	N of Rows in Working Data File	439
Missing Value Handling	Definition of Missing	User-defined missing values are treated as missing.
	Cases Used	Statistics for each analysis are based on cases with no missing data for any variable in the analysis.
Syntax		ONEWAY toptim BY marital /ES=OVERALL /STATISTICS DESCRIPTIVES HOMOGENEITY BROWNFORSYTHE WELCH /PLOT MEANS /MISSING ANALYSIS /CRITERIA=CILEVEL (0.95) /POSTHOC=TUKEY ...
Resources	Processor Time	00:00:00.11
	Elapsed Time	00:00:00.10

Descriptives

Total Optimism

	N	Mean	Std. Deviation	Std. Error	95% Confidence Interval for Mean Lower Bound
SINGLE	104	21.51	4.520	.443	20.63
STEADY RELATIONSHIP	37	21.59	4.524	.744	20.09
LIVING WITH PARTNER	36	21.64	4.549	.758	20.10
MARRIED FIRST TIME	188	22.51	4.199	.306	21.90
REMARRIED	29	23.00	4.488	.833	21.29
SEPARATED	10	18.50	4.994	1.579	14.93
DIVORCED	24	22.67	4.280	.874	20.86
WIDOWED	7	25.57	4.315	1.631	21.58
Total	435	22.12	4.429	.212	21.70

Descriptives

Total Optimism

	95% Confidence Interval for Mean Upper Bound	Minimum	Maximum
SINGLE	22.39	7	30
STEADY RELATIONSHIP	23.10	10	29
LIVING WITH PARTNER	23.18	8	29
MARRIED FIRST TIME	23.11	9	30
REMARRIED	24.71	15	29
SEPARATED	22.07	10	28
DIVORCED	24.47	15	30
WIDOWED	29.56	16	29
Total	22.53	7	30

Tests of Homogeneity of Variances

		Levene Statistic	df1	df2	Sig.
Total Optimism	Based on Mean	.264	7	427	.967
	Based on Median	.442	7	427	.876
	Based on Median and with adjusted df	.442	7	405.166	.876
	Based on trimmed mean	.334	7	427	.938

ANOVA

Total Optimism

	Sum of Squares	df	Mean Square	F	Sig.
Between Groups	329.264	7	47.038	2.454	.018
Within Groups	8183.757	427	19.166		
Total	8513.021	434			

ANOVA Effect Sizes^{a,b}

		Point Estimate	95% Confidence Interval	
			Lower	Upper
Total Optimism	Eta-squared	.039	.001	.065
	Epsilon-squared	.023	-.015	.050
	Omega-squared Fixed-effect	.023	-.015	.049
	Omega-squared Random-effect	.003	-.002	.007

a. Eta-squared and Epsilon-squared are estimated based on the fixed-effect model.

b. Negative but less biased estimates are retained, not rounded to zero.

Robust Tests of Equality of Means

Total Optimism

	Statistic ^a	df1	df2	Sig.
Welch	2.069	7	53.332	.063
Brown-Forsythe	2.318	7	118.374	.030

a. Asymptotically F distributed.

Post Hoc Tests

Multiple Comparisons

Dependent Variable: Total Optimism

Tukey HSD

(I) marital status	(J) marital status	Mean Difference (I-J)	Std. Error	Sig.	95% ... Lower Bound
SINGLE	STEADY RELATIONSHIP	-.085	.838	1.000	-2.64
	LIVING WITH PARTNER	-.129	.847	1.000	-2.71
	MARRIED FIRST TIME	-.996	.535	.579	-2.63
	REMARRIED	-1.490	.919	.737	-4.29
	SEPARATED	3.010	1.449	.432	-1.41
	DIVORCED	-1.157	.991	.941	-4.18
	WIDOWED	-4.062	1.709	.256	-9.27
STEADY RELATIONSHIP	SINGLE	.085	.838	1.000	-2.47
	LIVING WITH PARTNER	-.044	1.025	1.000	-3.17
	MARRIED FIRST TIME	-.911	.787	.943	-3.31
	REMARRIED	-1.405	1.086	.901	-4.71
	SEPARATED	3.095	1.560	.494	-1.66
	DIVORCED	-1.072	1.147	.983	-4.57
	WIDOWED	-3.977	1.804	.351	-9.47
LIVING WITH PARTNER	SINGLE	.129	.847	1.000	-2.45
	STEADY RELATIONSHIP	.044	1.025	1.000	-3.08
	MARRIED FIRST TIME	-.866	.796	.959	-3.29
	REMARRIED	-1.361	1.092	.918	-4.69
	SEPARATED	3.139	1.565	.479	-1.63
	DIVORCED	-1.028	1.154	.987	-4.54
	WIDOWED	-3.933	1.808	.369	-9.44
MARRIED FIRST TIME	SINGLE	.996	.535	.579	-.63
	STEADY RELATIONSHIP	.911	.787	.943	-1.49
	LIVING WITH PARTNER	.866	.796	.959	-1.56
	REMARRIED	-.495	.873	.999	-3.16
	SEPARATED	4.005	1.421	.093	-.32
	DIVORCED	-.161	.949	1.000	-3.05
	WIDOWED	-3.066	1.685	.607	-8.20
REMARRIED	SINGLE	1.490	.919	.737	-1.31
	STEADY RELATIONSHIP	1.405	1.086	.901	-1.90
	LIVING WITH PARTNER	1.361	1.092	.918	-1.97
	MARRIED FIRST TIME	.495	.873	.999	-2.17
	SEPARATED	4.500	1.605	.097	-.39
	DIVORCED	.333	1.208	1.000	-3.35
	WIDOWED	-2.571	1.844	.859	-8.19
SEPARATED	SINGLE	-3.010	1.449	.432	-7.42
	STEADY RELATIONSHIP	-3.095	1.560	.494	-7.85

Multiple Comparisons

Dependent Variable: Total Optimism

Tukey HSD

(I) marital status	(J) marital status	95% ... Upper Bound
SINGLE	STEADY RELATIONSHIP	2.47
	LIVING WITH PARTNER	2.45
	MARRIED FIRST TIME	.63
	REMARRIED	1.31
	SEPARATED	7.42
	DIVORCED	1.86
	WIDOWED	1.15
STEADY RELATIONSHIP	SINGLE	2.64
	LIVING WITH PARTNER	3.08
	MARRIED FIRST TIME	1.49
	REMARRIED	1.90
	SEPARATED	7.85
	DIVORCED	2.42
	WIDOWED	1.52
LIVING WITH PARTNER	SINGLE	2.71
	STEADY RELATIONSHIP	3.17
	MARRIED FIRST TIME	1.56
	REMARRIED	1.97
	SEPARATED	7.91
	DIVORCED	2.49
	WIDOWED	1.58
MARRIED FIRST TIME	SINGLE	2.63
	STEADY RELATIONSHIP	3.31
	LIVING WITH PARTNER	3.29
	REMARRIED	2.17
	SEPARATED	8.33
	DIVORCED	2.73
	WIDOWED	2.07
REMARRIED	SINGLE	4.29
	STEADY RELATIONSHIP	4.71
	LIVING WITH PARTNER	4.69
	MARRIED FIRST TIME	3.16
	SEPARATED	9.39
	DIVORCED	4.01
	WIDOWED	3.04
SEPARATED	SINGLE	1.41
	STEADY RELATIONSHIP	1.66

Multiple Comparisons

Dependent Variable: Total Optimism

Tukey HSD

(I) marital status	(J) marital status	Mean Difference (I-J)	Std. Error	Sig.	95% ... Lower Bound
	LIVING WITH PARTNER	-3.139	1.565	.479	-7.91
	MARRIED FIRST TIME	-4.005	1.421	.093	-8.33
	REMARRIED	-4.500	1.605	.097	-9.39
	DIVORCED	-4.167	1.648	.186	-9.19
	WIDOWED	-7.071 [*]	2.157	.025	-13.64
DIVORCED	SINGLE	1.157	.991	.941	-1.86
	STEADY RELATIONSHIP	1.072	1.147	.983	-2.42
	LIVING WITH PARTNER	1.028	1.154	.987	-2.49
	MARRIED FIRST TIME	.161	.949	1.000	-2.73
	REMARRIED	-.333	1.208	1.000	-4.01
	SEPARATED	4.167	1.648	.186	-.85
	WIDOWED	-2.905	1.881	.783	-8.63
WIDOWED	SINGLE	4.062	1.709	.256	-1.15
	STEADY RELATIONSHIP	3.977	1.804	.351	-1.52
	LIVING WITH PARTNER	3.933	1.808	.369	-1.58
	MARRIED FIRST TIME	3.066	1.685	.607	-2.07
	REMARRIED	2.571	1.844	.859	-3.04
	SEPARATED	7.071 [*]	2.157	.025	.50
	DIVORCED	2.905	1.881	.783	-2.82

Multiple Comparisons

Dependent Variable: Total Optimism

Tukey HSD

(I) marital status	(J) marital status	95% ... Upper Bound
	LIVING WITH PARTNER	1.63
	MARRIED FIRST TIME	.32
	REMARRIED	.39
	DIVORCED	.85
	WIDOWED	-.50
DIVORCED	SINGLE	4.18
	STEADY RELATIONSHIP	4.57
	LIVING WITH PARTNER	4.54
	MARRIED FIRST TIME	3.05
	REMARRIED	3.35
	SEPARATED	9.19
	WIDOWED	2.82
WIDOWED	SINGLE	9.27
	STEADY RELATIONSHIP	9.47
	LIVING WITH PARTNER	9.44
	MARRIED FIRST TIME	8.20
	REMARRIED	8.19
	SEPARATED	13.64
	DIVORCED	8.63

*. The mean difference is significant at the 0.05 level.

Homogeneous Subsets

Total Optimism

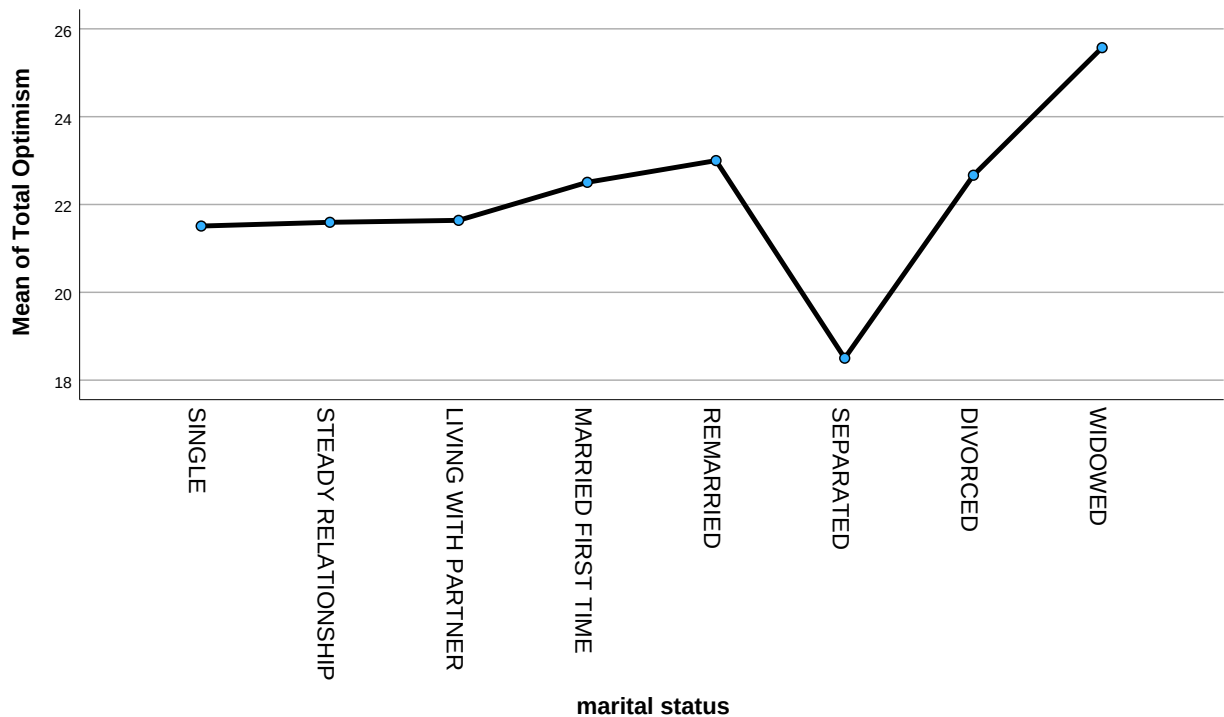
Tukey HSD^{a,b}

marital status	N	Subset for alpha = 0.05	
		1	2
SEPARATED	10	18.50	
SINGLE	104	21.51	21.51
STEADY RELATIONSHIP	37	21.59	21.59
LIVING WITH PARTNER	36	21.64	21.64
MARRIED FIRST TIME	188	22.51	22.51
DIVORCED	24		22.67
REMARRIED	29		23.00
WIDOWED	7		25.57
Sig.		.068	.061

Means for groups in homogeneous subsets are displayed.

- Uses Harmonic Mean Sample Size = 20.579.
- The group sizes are unequal. The harmonic mean of the group sizes is used. Type I error levels are not ...

Means Plots



Question3: Age Group and smoking on Optimism: Two-way Anova

Notes

Output Created		06-NOV-2024 19:00:44
Comments		
Input	Data	D:\survey.sav
	Active Dataset	DataSet2
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	Split File	<none>
	N of Rows in Working Data File	439
Missing Value Handling	Definition of Missing	User-defined missing values are treated as missing.
	Cases Used	Statistics are based on all cases with valid data for all variables in the model.
Syntax		UNIANOVA toptim BY agegp3 smoke /METHOD=SSTYPE(3) /INTERCEPT=INCLUDE /POSTHOC=agegp3 (TUKEY) /PLOT=PROFILE (agegp3*smoke) TYPE=LINE ERRORBAR=NO MEANREFERENCE=NO YAXIS=AUTO /PRINT ETASQ DESCRIPTIVE HOMOGENEITY /CRITERIA=ALPHA(.05) /DESIGN=agegp3 smoke agegp3*smoke.
Resources	Processor Time	00:00:00.16
	Elapsed Time	00:00:00.17

Between-Subjects Factors

		Value Label	N
age 3 groups	1	18 - 29	146
	2	30 - 44	153
	3	45+	133
smoker	1	YES	85
	2	NO	347

Descriptive Statistics

Dependent Variable: Total Optimism

age 3 groups	smoker	Mean	Std. Deviation	N
18 - 29	YES	20.50	5.550	34
	NO	21.63	4.217	112
	Total	21.36	4.567	146
30 - 44	YES	22.29	3.834	34
	NO	22.05	4.246	119
	Total	22.10	4.147	153
45+	YES	21.94	4.815	17
	NO	23.17	4.436	116
	Total	23.02	4.486	133
Total	YES	21.51	4.795	85
	NO	22.29	4.338	347
	Total	22.13	4.437	432

Levene's Test of Equality of Error Variances^{a,b}

		Levene Statistic	df1	df2	Sig.
Total Optimism	Based on Mean	2.085	5	426	.066
	Based on Median	1.459	5	426	.202
	Based on Median and with adjusted df	1.459	5	384.458	.202
	Based on trimmed mean	2.038	5	426	.072

Tests the null hypothesis that the error variance of the dependent variable is equal across groups.

a. Dependent variable: Total Optimism

b. Design: Intercept + agegp3 + smoke + agegp3 * smoke

Tests of Between-Subjects Effects

Dependent Variable: Total Optimism

Source	Type III Sum of Squares	df	Mean Square	F	Sig.	Partial Eta Squared
Corrected Model	247.214 ^a	5	49.443	2.557	.027	.029
Intercept	120572.131	1	120572.131	6235.733	<.001	.936
agegp3	105.173	2	52.587	2.720	.067	.013
smoke	31.078	1	31.078	1.607	.206	.004
agegp3 * smoke	31.913	2	15.956	.825	.439	.004
Error	8236.999	426	19.336			
Total	220132.000	432				
Corrected Total	8484.213	431				

a. R Squared = .029 (Adjusted R Squared = .018)

Post Hoc Tests

age 3 groups

Multiple Comparisons

Dependent Variable: Total Optimism

Tukey HSD

(I) age 3 groups	(J) age 3 groups	Mean Difference (I-J)	Std. Error	Sig.	95% Confidence Interval	
					Lower Bound	Upper Bound
18 - 29	30 - 44	-.74	.509	.313	-1.94	.45
	45+	-1.65*	.527	.005	-2.89	-.41
30 - 44	18 - 29	.74	.509	.313	-.45	1.94
	45+	-.91	.521	.189	-2.14	.32
45+	18 - 29	1.65*	.527	.005	.41	2.89
	30 - 44	.91	.521	.189	-.32	2.14

Based on observed means.

The error term is Mean Square(Error) = 19.336.

*. The mean difference is significant at the .05 level.

Homogeneous Subsets

Total Optimism

Tukey HSD^{a,b,c}

age 3 groups	N	Subset	
		1	2
18 - 29	146	21.36	
30 - 44	153	22.10	22.10
45+	133		23.02
Sig.		.327	.187

Means for groups in homogeneous subsets are displayed.

Based on observed means.

The error term is Mean Square(Error) = 19.336.

- Uses Harmonic Mean Sample Size = 143.513.
- The group sizes are unequal. The harmonic mean of the group sizes is used. Type I error levels are not guaranteed.
- Alpha = .05.

Profile Plots

